Functional Annotation of *hmgA* (Homogentisate 1,2-Dioxygenase, Q88E47) in *Pseudomonas putida* KT2440

0. Identity verification (performed before research)

The target was confirmed to be the correct protein before any literature was interpreted:

- **UniProt:** Q88E47 = **HGD_PSEPK** (reviewed).
- **Gene / locus:** `hmgA` / `PP_4621`.
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950).
- **Length / mass:** 433 aa, ~48 kDa.
- **EC 1.13.11.5**, homogentisate dioxygenase family (Pfam/InterPro HgmA_N + HgmA_C; RmlC-like jelly-roll / cupin-type core, IPR014710).
- **Decisive check:** UniProt Q88E47 cross-references PDB entries **3ZDS, 4AQ2, 4AQ6**, which are the structures reported in **Jeoung et al., 2013 (** P 23858455 **)**. The iron-ligand residues annotated for Q88E47 (His331, Glu337, His367), the substrate-binding Tyr346, and the catalytic His288 match that paper's numbering exactly. **Therefore the 2013 Dobbek-group structural/mechanistic study is of this exact protein, not a homolog** — giving direct experimental evidence for the target.

No ambiguity: the gene symbol, organism, family, and domains all align with a well-characterized enzyme.

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1. Summary (answer to the research question)

HmgA is homogentisate 1,2-dioxygenase (EC 1.13.11.5), a cytoplasmic, mononuclear **non-heme Fe(II)** ring-cleaving oxygenase. It catalyzes the committed, central step of the **homogentisate pathway** by using molecular O_2 to oxidatively cleave the aromatic ring of

homogentisate (2,5-dihydroxyphenylacetate), producing **4-maleylacetoacetate**. This reaction channels carbon derived from **L-phenylalanine and L-tyrosine** (and 3-hydroxyphenylacetate) into central metabolism, ultimately yielding **fumarate + acetoacetate**. It acts inside the cell (soluble cytoplasmic enzyme) and its expression is induced by its own substrate homogentisate and repressed by the IclR-type regulator HmgR.

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2. Primary molecular function: the catalyzed reaction and substrate specificity

Reaction:

> homogentisate (2,5-dihydroxyphenylacetate) + O_2 → 4-maleylacetoacetate

HmgA is a **dioxygenase**: both atoms of O_2 are incorporated into the product as it opens the benzene ring between C1 and C2 of homogentisate. The enzyme is highly specific for homogentisate, a 1,2,4-trisubstituted aromatic (a hydroquinone bearing an acetate side chain), which is the natural central intermediate of aromatic amino-acid breakdown (P 23858455; P 15262943).

Cofactor: a single catalytic **Fe(II)** ion (non-heme). The 1.70 Å resting-state structure of the *P. putida* enzyme shows the iron in **octahedral coordination by two histidines (His331, His367), a bidentate carboxylate (Glu337), and two water molecules** — a variant of the classic 2-His-1-carboxylate "facial triad" used by many Fe(II)/O_2 oxygenases (P 23858455).

Mechanistic evidence (direct, on this protein): Jeoung et al. trapped and crystallographically resolved **four consecutive states of the catalytic cycle** in the *P. putida* enzyme: the **substrate-bound**, **superoxo:semiquinone-**, **alkylperoxo-**, and **product-bound** intermediates. Homogentisate binds as a **monodentate** Fe ligand; its interaction with **Tyr346** triggers folding of a mobile loop that closes over the active site and excludes solvent, and substrate binding is enthalpy-driven/entropically disfavored (anoxic ITC). This shows that, despite an unrelated fold, HmgA proceeds through the **same principal O_2-activation intermediates as extradiol ring-cleaving dioxygenases** (P 23858455).

Residue function (mutagenesis on this protein, curated in UniProt from P 23858455**):**
- **His288 (active-site proton acceptor):** H288F reduces catalytic efficiency ~**75-fold**. - **Tyr346 (substrate anchor):** Y346F decreases homogentisate affinity >**60-fold** and catalytic efficiency ~**20-fold**.

Quantum-chemical support: QM/MM (P 27119315) and hybrid-DFT (P 16332080) studies of homogentisate dioxygenase reconstruct a mechanism of **peroxo-intermediate formation → O–O bond cleavage (arene-oxide radical) → ring scission**, with an active-site glutamate stabilizing bridging species — consistent with the crystallographically observed alkylperoxo intermediate. These computational models rationalize the enzyme's product specificity.

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3. Biological process and pathway context

HmgA is the **first (ring-opening) enzyme of the "central" homogentisate catabolic pathway** in *P. putida* (P 15262943):

Peripheral (upstream) reactions feeding homogentisate: - **PhhAB** — phenylalanine hydroxylase: L-Phe → L-Tyr - **TyrB** — aminotransferase: L-Tyr → 4-hydroxyphenylpyruvate (4-HPP) - **Hpd** — 4-HPP dioxygenase: 4-HPP → **homogentisate**

Central pathway (the hmgABC operon): 1. **HmgA** (this protein): homogentisate → **4-maleylacetoacetate** 2. **HmgC** — maleylacetoacetate isomerase: 4-maleylacetoacetate → 4-fumarylacetoacetate 3. **HmgB** — fumarylacetoacetate hydrolase: 4-fumarylacetoacetate → **fumarate** + **acetoacetate**

The end products (fumarate, a TCA-cycle intermediate, and acetoacetate) let *P. putida* use Phe/Tyr — and, via the same route, **3-hydroxyphenylacetate** — as carbon and energy sources. An engineered cassette carrying *hmgABC*, *hpd* and *tyrB* confers tyrosine utilization on heterologous bacteria, demonstrating the pathway's sufficiency (P 15262943).

Regulation: The *hmgABC* genes form a **single transcriptional unit**. The divergently transcribed gene **hmgR** encodes an **IclR-type repressor**; **homogentisate itself is the inducer**, relieving repression when substrate is present. HmgR was reported as the **first IclR-type regulator shown to act as a repressor of an aromatic catabolic pathway**, protecting a 17-bp palindrome/direct-repeat region of the P*hmg* promoter (P 15262943).

Physiological consequence of loss of function: Blocking HmgA causes homogentisate to accumulate; homogentisate auto-oxidizes and polymerizes into the brown pigment **pyomelanin** (ochronotic pigment). This is the bacterial counterpart of the human disease **alkaptonuria**, caused by deficiency of the orthologous HGD/HGO enzyme (P 9244427; and analogous plant HGO loss browns soybean seed, P 27660165) — a cross-kingdom confirmation of the conserved committed-step role.

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4. Localization

HmgA is a **soluble cytoplasmic enzyme**. It has no signal peptide, no transmembrane segments, and no lipidation/secretion signals in Q88E47; its substrate (homogentisate) and downstream enzymes (HmgB, HmgC) are intracellular products of cytoplasmic amino-acid catabolism. The crystallized protein is a soluble multimer (the crystallographic asymmetric unit of 3ZDS/4AQ2/4AQ6 contains 12 chains; HGDO enzymes assemble as homo-oligomers, and the human ortholog is a hexamer arranged as a dimer of trimers). Thus HmgA carries out its function **in the cytoplasm**.

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5. Structural fold

HmgA adopts a **cupin/RmlC-like β-jelly-roll core** (InterPro IPR014710) that houses the mononuclear Fe(II) site, flanked by the family-specific HgmA N- and C-domains (IPR046452/IPR046451). This fold is distinct from both catechol-cleaving intradiol and classic extradiol dioxygenases, yet supports an analogous Fe(II)/O_2 chemistry — an example of **convergent catalytic strategy** on divergent scaffolds (P 23858455).

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6. Hypotheses: supported vs. refuted

Supported - HmgA catalyzes homogentisate + O_2 → 4-maleylacetoacetate (direct: trapped substrate- and product-bound structures of this protein; P 23858455). - Catalysis uses a mononuclear non-heme Fe(II) center coordinated by His331/His367/Glu337 (direct crystallography + matching UniProt annotation). - HmgA is the committed central step of Phe/Tyr catabolism via the *hmgABC* operon, regulated by HmgR/homogentisate (P 15262943). - Enzyme is cytoplasmic (sequence features; substrate/pathway context).

Refuted / ruled out - HmgA is **not** a heme enzyme, **not** a monooxygenase, and **not** membrane-bound. - Gene-symbol ambiguity was ruled out: the "hmgA" and PDB evidence map unambiguously to Q88E47/PP_4621 (no mis-identification with unrelated "HMG" genes such as HMG-CoA reductase or HMGB chromatin proteins).

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7. Limitations and future directions

- Kinetic parameters (kcap/Km) for the wild-type *P. putida* enzyme are referenced through mutagenesis fold-changes; a full steady-state kinetic table for KT2440 HmgA would sharpen the substrate-specificity claim.
- The precise biological oligomeric state (hexamer vs. other) of the *P. putida* enzyme should be confirmed by solution methods (SEC-MALS).
- Whether HmgA has measurable activity toward homogentisate analogs (substrate-specificity breadth) is not fully mapped experimentally.

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8. Key references

- Jeoung, Bommer, Lin, Dobbek (2013) *PNAS* — crystal structures/intermediates of this enzyme.

  P 23858455

  (PDB 3ZDS/4AQ2/4AQ6)
- Arias-Barrau et al. (2004) *J. Bacteriol.* — homogentisate pathway & hmgABC/hmgR regulation in *P. putida*.

  P 15262943
- Qi, Lu, Lai (2016) — QM/MM mechanism.

  P 27119315
- Borowski, Georgiev, Siegbahn (2005) — DFT mechanism.

  P 16332080
- Granadino et al. (1997) — human HGO gene / alkaptonuria (ortholog).

  P 9244427
- Stacey et al. (2016) — plant HGO loss-of-function phenotype (ortholog).

  P 27660165